HOLONOMIC HYBRID SOURCE SEEKING

Two overlapping regions, drawn in the physical workspace

Computed level sets and averaged flows · identical source, obstacle and initial position

q = 1 Region U_1

3D CONSTRUCTION

The left-side cover connects the start to the source.



Solid boundary: ∂U_1 Dashed diamond: excluded inner set

Red dashes: $\Gamma(p, 1) = \delta \rightarrow$ select $q^+ = 2$; position stays fixed.

q = 2 Region U_2

3D CONSTRUCTION

The right-side cover gives a second route to the source.



Solid boundary: ∂U_2 Dashed diamond: excluded inner set

Red dashes: $\Gamma(p, 2) = \delta \rightarrow$ select $q^+ = 1$; position stays fixed.

Blue arrows: averaged flow Red dashes: switching threshold White curves: potential level sets

q selects V_q. At a jump, only q changes: $p^+ = p$. The two averaged routes shown retain their initial q.

$\rho = 0.92 \cdot \rho_b \approx 0.73820 \cdot \delta = 0.15 \cdot k = 0.05 \cdot$ source (0, 2.4) · start (0, −2)